The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Abstract and storyline

Abstract

Temporal-interference (TI) stimulation can act on neural populations only if three stages work in series: the kilohertz field must reach a nonlinear membrane (access), the membrane must generate the difference frequency absent from the field (detection), and the circuit must amplify that weak component (amplification). We build a mesoscale model of all three stages. Following single-neuron TI models that locate rectification in nodes and the axon initial segment, we represent access by a short effective membrane time. We analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire neurons, and its spiking network. For weak fields the detector is quadratic: the beat is proportional to the product of the local carrier amplitudes, and a signed steady source to their summed squares. A static rate curve and the exact mean field give the same product law; the mean field adds frequency dependence when the membrane is slow relative to the input. Near a Hopf bifurcation the circuit amplifies the beat as the inverse of the modal damping; with the Jansen–Rit operating point held fixed, synaptic coupling sets this gain independently of detection. The amplified response appears mainly in envelope-locked timing, with little change in mean rate. At stronger fields the steady shift alone can drive a gamma circuit across its oscillation threshold, and an unmodulated carrier of equal power does the same. The model expresses TI efficacy as the product of three separately measurable factors. For weak fields, dose follows the product of carrier amplitudes rather than the envelope minimum. Under uniform conductivity, access, and detection, the steady drive has its largest magnitude at compartment boundaries. TI shares its circuit amplifier with direct low-frequency stimulation. Carrier sweeps, independent-amplitude tests, beat sweeps across states, and equal-power controls each test one stage.

The article, technical note, and storyline share the four abstract macros in `planning/tn0484_abstract.tex`. The JNE article is the working edition; only its run-in headings differ. Draft10 implements the placement decisions relayed on 5 October: LaNMM methods join their results in Appendix E; transduction laws remain in Theory, with their measurement procedure in Methods; the equal-RMS carrier control moves from S13 to the main article (now Figure 13). All scientific content is retained. Draft11 corrects notation, citation scope, and the access argument, and integrates the critical-mode and genuine two-tone checks. Draft12 presents the mesoscale model as access, detection, and amplification in series, each with its own test, in the shared abstract and the Introduction's closing paragraph. The closing record gives the reasons for the placements.

Revision approved from the 6 October review copy. The envelope oscillation and its spatial map become a separate Theory subsection; the mean shift and the two mean-map restrictions follow in another. JR results stay together, followed by the NMM2 mean field, its coupling sweep, and the QIF network. Predictions move to the Discussion. The approved model schematic follows the model definitions. Numerical qualifications are retained; the QIF cutoff diagnostics move to Methods so its Results can lead with the distinction between beat modulation and induced gamma. The reading redline remains against draft13.

Approved readability revision. This pass gives TI versus tACS its own Results subsection after the JR chain; NMM2 and QIF follow. The AM-surrogate protocol is in Methods. The compartment matrix, QIF derivation, and cable estimate are in the model appendix; their results remain in Theory. The eight-state PING comparison belongs with the NMM2 Results. Predictions are consolidated in their Discussion table with each test retained. Passages marked CUT? remain pending an explicit decision. The Conclusion states the product/DC law, shared amplifier, timing response, and carrier control. Numerical scope and the fixed draft13 redline are retained.

Problem and theory

First argument block: Sections 1–2

A two-tone field carries no power at its beat frequency. Access converts the applied field into local membrane polarization; nonlinear detection generates the beat; the network selects and amplifies that generated input. The Introduction develops these stages in that order, with their mechanisms and evidence. The mesoscale model represents the stages in series, with carrier-frequency sweeps of membrane polarization testing access, independent-amplitude sweeps testing detection, and beat-frequency sweeps across operating states testing amplification. Equal-power carrier controls separate the steady shift from beat-locked activity. Axonal TI rectification is established in prior models; short effective membrane times were already tested in a TI gamma-network model (Esmaeilpour et al., 2021). Our contribution is the separation of mixing from recurrent gain, the exact QIF mean field, the product law, and the shared TI/tACS amplifier. Fast axonal compartments are the access hypothesis: Cohen's periaxonal circuit supports distinct nodal and internodal voltage dynamics; the AIS is a separate candidate with a high sodium-channel density. Neither observation fixes a TI charging time. Luff supplies evidence for cellular mixing, while cellular resonance and recurrent amplification remain distinct. QIF connects the effective weak-input account to a neuron model; its static transfer is an adiabatic limit, not a consequence of small amplitude alone.

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§1 Introduction	The two-tone field (Eq. 1) has lines only at the applied tones; a linear system cannot create the beat. Cellular mixing exists (MRG axon model with Hodgkin–Huxley dynamics; Luff's subthreshold mixer at 5 kHz, TTX-sensitive). Primate timing changes establish modulation; synaptic blockade and network simulations support circuit involvement. Retain weak modulation of ongoing activity as the scope, with direct stimulation thresholds and block outside it. Weak input can perturb spiking neurons; linear gain requires stability under the rectified mean shift. Introduce fast axonal access before detection: AIS and nodes are candidates; Cohen's submyelin circuit explains distinct axonal voltage modes, not a measured TI transfer. Cellular contributions enter an effective quadratic response, and cellular tuning remains distinct from the recurrent amplifier. Retain the candidate positively and name its calibration gap.	States the physical sequence: field-to-membrane access, beat generation, then amplification. Places each reference beside the stage it supports and closes with a separate test for each. Announces the ladder (JR column, QIF mean field, spiking network, laminar column) once. The transfer curve must be defined before its curvature can mix, so §2.1 follows.	Fig. 1 (receiver concept). Cohen 2020; Kole 2008 (AIS); Mirzakhalili 2020; Luff 2024 (mixer); Vieira 2024; Caldas-Martinez 2024; CFC and HAM companion papers.

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§2.1 Population transfer and the slow-synapse limit	JR synaptic filter and sigmoid (Eqs. 2–3); NMM1 with the direct field term $\Lambda[E]$ (Eq. 3). The MPR mean field (Eq. 4) and its static curve $F(Q)$ (Eq. 5), inflection at $Q = \Delta/\sqrt{3}$, nonlinear even at zero recurrence and unbounded. Retain threshold recruitment here and the single-cell square-root rate in App. B.2. The slow-synapse reduction (Eq. 6) and its two conditions: $\tau_{\mathrm{rel}} \ll \tau_s$ makes the transfer static for synaptic input; $\omega_c \tau_{\mathrm{rel}} \ll 1$ is the stronger condition for a carrier. Effective curvature collects static mixing; the coefficient's value and dynamics remain model-specific. Cellular and circuit resonance are distinct; the reduction removes membrane-dependent modes.	Defines what a transfer curve is and when it is static, so that curvature can be a mixer. Introduces the time-scale distinction that the Discussion and Conclusion reuse. The field must reach the curve, which §2.2 supplies.	Fig. 2 (model schematic); Rosetta Stone (Ruffini 2025); Montbrió 2015; Clusella 2023; de Palma Aristides 2026 (WP0242, unpublished); Hutcheon and Yarom 2000.
§2.2 From applied field to a squared-envelope beat	Quasistatic field; polarization $s(t)$ through parallel entry points with access filters (Eqs. 7–8), one compartment response counted once. Taylor expansion (Eqs. 9–10) with the truncation condition $\lvert s\rvert \ll 1/\rho \approx 1.8$ mV. The squared-envelope map is derived in §2.5; the AM-surrogate protocol is in §3.1. The effective dynamic coefficient $\chi^{(2)}$ combines cellular contributions and reduces to $\Sigma''/2$ in the static case. Retain smoothness and nonzero second-order response, its possible frequency dependence, and the DC term. Static predictions: ε^2 scaling, Σ'' law, carrier independence at fixed polarization.	Derives the beat and its amplitude: the first result. Fixes the waveform discipline (two tones for every waveform-sensitive claim). The derivation assumes a static curve, so §2.3 asks what a dynamic population does.	Figs. 4 and 7 (two-tone runs); Fig. 7d (surrogate check). HAM paper (Volterra kernel); Galan and Gadea 2023.
§2.3 The exact mean field connects static and dynamic mixing	Single-carrier DC coefficient $\kappa_{\mathrm{DC}}(Q, \tilde{\omega})$ (Eq. 11), with limits $F''/4$ and $F'/(2\tilde{\omega}^2)$ (Eq. 12), sign change at $\tilde{\omega}_z$ above the inflection. Fixed-excursion coefficient $\kappa_V(Q, \tilde{\omega})$ (Eq. 13): the $1/\tilde{\omega}^2$ is the membrane, not a detector filter. The Jacobian and phasor derivation is in App. B.2; numerical agreement with the full PING response is in §4.9. Carrier and beat take different routes to the rate (Table 1).	Connects the static and dynamic QIF descriptions, with membrane access counted once. Retain the distinction between the DC calculation and the two-tone beat source. Whatever the detector, its envelope law has observable signatures, which §2.4 derives.	Fig. 10; Table 1. Code: `static_vs_dynamic_check.py`, `mpr_twotone_lines_check.py`.
§2.4 Detector laws and harmonic signatures	Cycle average (Eq. 14) is even in $\mathcal{A}$; nonzero curvature gives a leading square law for a smooth memoryless transfer. Specified ideal rectifiers follow $\mathcal{A}$; a step detector follows neither (harmonic ratio 0.151). Harmonic ladder 1/5, 3/35, 1/21 and $\mathcal{H}_{\mathrm{det}}$ (Eq. 15), whose inverse needs a sign or branch choice. The square-law harmonic ratio vanishes at weak input; a threshold detector has an onset and an amplitude-dependent ratio. The network weights each harmonic by its gain.	Separates detector classes by amplitude scaling and harmonics. Their different spatial predictions follow in a dedicated subsection, rather than being compressed into this one.	App. D; Fig. 7b (slope 1.99); measurement procedure in §3.2.

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§2.5 Envelope oscillations and spatial weighting	Spatially varying signed polarizations $\varepsilon_j(\mathbf{x})$ and their nonnegative envelope and its square (Eq. 16): the square is sinusoidal, the envelope is a rectified cosine at matched tones. Table 2 compares slow signals, means, and excursions before detector gains, then gives the signed steady sources after the specified gains. Excursion map (Eq. 17): the quadratic product weights the envelope minimum by the larger carrier. Retain the 2× versus 40× two-site example and its geometric limits. Both beat maps can peak centrally; Fig. S13a is a line profile, not a head model. The real-montage finite-element comparison in §4.5 tests this geometry; a physiological beat-map comparison remains a prediction. Spatial response (Eq. 18) includes curvature, network gain, and signed vector coupling. With fixed pairs, frequencies, access, orientations, and neural state, the current ratio only rescales the quadratic beat map. Steering requires a change in field shape or response regime; the signed steady source can change state indirectly.	Explains how the same applied field yields different dose maps under the two detector laws. The product need not preserve envelope-map contrast, and network tuning adds dynamical selectivity. The mean component requires its own spatial analysis.	Table 2; Fig. S13a; Grossman 2017; Cassarà et al. 2025, Part II, §5.7.
§2.6 The steady source has a state-dependent sign	Defines the envelope mean and mean-square polarization, with matched-tone values $4\varepsilon/\pi$ and ε^2, before detector gain. The applied field has no DC. The JR source is $Y_{\mathrm{DC}} = \Sigma''\mathcal{I}/2$: positive, negative, or zero according to curvature; the circuit determines the subsequent mean response. Orientation reversal leaves the squares unchanged. Fast-carrier QIF gives a positive input shift, whereas its adiabatic rate shift follows signed F''. The geometric mean-square map sums intensities after beat averaging: 202× superficial/deep contrast in the example and local beat/DC at most one. Both idealized mean maps are subharmonic; the envelope is a norm of harmonic components at each beat phase. Retain the homogeneous, isotropic, source-free and fixed-coupling assumptions, the distinction from weighted physiological response, and cancellation of the beat for orthogonal fields under isotropic averaging. Equal-RMS mean-shift equivalence requires equal second-order DC coefficients across the carrier interval.	Restricts mean-map focusing without excluding an interior beat maximum. Conductivity interfaces are compartment boundaries. Variable access, curvature, or orientation can change the physiological map; adaptation can change the mean response. A uniform negative coefficient puts the most negative source at the boundary, where its magnitude is largest.	Fig. S13b,c; Table 2; equal-RMS control in §4.10.
§2.7 Amplification: synaptic low-pass and network resonance	Synaptic transfer H_s (16 Hz corner; 0.7 at 10 Hz, 2.5% at 100 Hz). One-mode reduction, $\mu_\pm = -\gamma \pm i\omega_0$, $\mathcal{Q} = \omega_0/2\gamma$, gain $\propto 1/\gamma$ (Eq. 19); entrainment above the Hopf. Receiver equation (Eq. 20) with D and G and their normalization for JR and QIF. The cubic normal form has no quadratic mixer.	Supplies the second factor of the receiver and the claim that a large susceptibility is not evidence of detection. The carrier must still reach the mixer, which §2.8 quantifies.	Fig. 6; Figs. 5 and 9; App. C.

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§2.8 Carrier access depends on the effective membrane response	Point-membrane H_m (Eq. 21); $\lvert H_m\rvert^2 \approx 10^{-4}$ at 1 kHz for 16 ms, 0.39 and 0.14 at 1 and 2 kHz for 0.2 ms, labeled as assumed. Table 3 separates somatic measurements, Cohen's approximately 1 ms minimum axonal decay, assumed fast access, prior 1/0.4 ms TI network parameters, channel kinetics, and synaptic filtering. Cohen is cited here as well as in the Introduction and Discussion. The cable derivation (Eq. 29) is in App. B.1; its sealed-end quadratic suppression of about 10^{-2} at 1 kHz remains here. Channel kinetics belong in the effective mixing response, counted once; no universal gate cutoff follows from QIF. Carrier sweeps constrain access plus mixing; beat sweeps probe the amplifier. The plotted simulations retain finite membrane times; fast QIF relaxation is an analytic limit.	Separates carrier independence at fixed polarization from the applied-field roll-off and connects the fast-compartment hypothesis to a reduced-model sensitivity calculation. The detector coefficient also depends on state, which §2.9 takes up.	Figs. S5, S6, S9; Deitcher 2017; Eyal 2016; Cohen 2020; Discussion comparison with Luff and Semenov.
§2.9 Operating-point dependence	Signed PEIX (Eq. 22) is zero at the inflection and approaches ±1 in saturation; the normalized slope is $1 - \lvert\text{PEIX}\rvert$. Static mixing vanishes at the inflection; slope and curvature play distinct roles in the memoryless detector. The fast-carrier QIF DC coefficient follows F'; a beat also needs its finite-frequency susceptibility. State shifts scale, and in JR can flip, the response. Retain the slope-versus-curvature distinction.	Turns the detector coefficient into a brain-state prediction and marks which part is model-specific. The theory is complete; Methods specify the simulations that test it.	Fig. 3b; Fig. 7a; CFC paper (PEIX).

JR, Jansen–Rit; MRG, McIntyre–Richardson–Grill; QIF, quadratic integrate-and-fire; MPR, Montbrió–Pazó–Roxin; NMM1/NMM2, sigmoid and exact mean-field neural mass models; PING, pyramidal–interneuron network gamma; AM, amplitude modulation; PEIX, Population Excitation–Inhibition Index; CFC, cross-frequency coupling; HAM, hierarchical amplitude modulation.

Simulations

Second argument block: Sections 3–4

The JR column tests the predictions of §2.2, and the exact mean field derives generation and amplification from QIF dynamics. Finite-amplitude spiking simulations illustrate the coexistence of beat-locked and gamma activity. Methods stand apart because the lock-in read-out and the pinned-operating-point protocol recur across models.

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§3.1 Waveforms and matched-source controls	AM surrogate (Eq. 23): one central line and two sidebands, with the same leading beat as two tones but a 3/4 DC source and an extra second harmonic at $m = 1$. Retain waveform-specific uses, open-loop control, and two-tone reruns of mean-shift-dependent comparisons.	States the protocol after the detector laws; DC and harmonic claims cannot rely on leading-beat equivalence alone.	Fig. 7d; §4.7–4.9.
§3.2 Amplitude scaling and harmonic measurements	Lock-in (Eq. 24): Hann window, weighted-mean subtraction, physical-amplitude normalization, and finite-window leakage limits. Amplitude-slope test at fixed coupling and linear downstream gain. Harmonic ratio corrected by $\lvert G(2\Omega)/G(\Omega)\rvert$: at resonance and $\mathcal{Q} = 10$, 20% reads 0.7%. Off resonance at $\Delta f \simeq f_0/4$, the correction is 1.22–1.25; an independent weak tACS sweep provides the alternative deconvolution.	Collects the measurement procedure after its theoretical predictions, preserving the estimator and the restrictions on interpreting a measured harmonic ratio.	Fig. 7b; §2.4; §4.4 and §5.1.
§3.3 Head model and field computation	IXI002, SimNIBS head model, FreeSurfer hippocampal label, and the Violante montage; linear, nondispersive conductivity and uniform polarization coupling. The direction maximizing the envelope at 1:1 stays fixed across current ratios. Retain the isotropic signed-mixing average, raw and square-root contrasts, node-based percentile and focus definitions, and the 32-condition sensitivity grid. Geometric maps do not supply detector coefficients or a circuit response.	Defines the anatomical model and comparisons needed to evaluate the product law without changing orientation as the currents change. This makes the spatial result in §4.5 reproducible from the supplied FEM outputs.	Results §4.5 and Fig. 8; `docs/fem_ixi002_violante.md`.
§3.4 Jansen–Rit: integration, lock-in, controls	Field enters only the pyramidal sigmoid; the read-out $v = y_1 - y_2$ is not fed the field. RK4 steps with $< 1\%$ convergence; settling and windows. Upper Hopf at $p \approx 315.70$, $f_0 \approx 11.16$ Hz; alpha cycle between Hopf points near 90 and 316. The SNIC near 114 belongs to the larger cycle. The estimator in §3.2 retains DC subtraction to 10^{-8}. Open-loop detector versus closed-loop column; square law, linearization, finite-difference checks.	Specifies the JR integrator and lock-in; run-specific reproduction limits remain recorded.	Fig. 3; Grimbert and Faugeras 2006; `run_all.py`.

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§3.5–3.6 NMM2 PING and QIF network	PING (Eq. 25): weights $(1,1,1,2)$, $\bar{\eta}_H = 1.110$, $f_0 = 54$ Hz, field as current $I_E = \lambda_I E$ so the membrane computes access with $\tau_m = \tau_E$; two criticality knobs. Critical-mode continuation and a 15–90 Hz linear-response grid for the coupling sweep; separate normalizations for the plotted gain and the $A_f = 2, 4, 8$ amplitude check. QIF: $N = 8000$ per population, Lorentzian quantiles, beats 55 and 42 Hz at $\bar{\eta} = 0.85/1.5$, defined by the field-free states; $A_f = 35$ is finite amplitude, with carrier excursion about 1.2 per tone. Modulation depth and pairwise phase consistency; the 45% threshold sensitivity of phase concentration.	Fixes the models whose results follow and states which comparisons are perturbative. The LaNMM method now accompanies its results in App. E.1.	Eq. 25; Rosetta Stone PING parameters; Vinck 2010.
§3.7–3.8 Coupling sweeps, tACS control, vocabulary	Pinning: solve $p(C)$ for fixed v^*, read γ, ω_0 from the Jacobian; tACS control at $\varepsilon = 0.1$ mV. The JR coupling figure uses the AM surrogate at 0.3 mV and $m = 1$; the timing figure uses genuine two tones. TI/tACS and endpoint controls test both waveforms. Pinning is field-free and does not cancel driven mean shifts. Forced response, phase capture, and amplitude gating are distinct. The maps measure frequency dominance ($\lvert f_{\text{out}} - \Delta f \rvert < 0.3$ Hz); the rate fold does not uniquely identify gamma-amplitude gating.	The pinning protocol holds the local detector curvature fixed; the vocabulary keeps "entrainment" from covering gating.	App. F (metrics); Pikovsky 2001; Canolty 2010; Tort 2010.
§4.1–4.2 The sigmoid demodulates; the beat drives a resonance	An alpha line from two tones with no alpha power in the input. Approximately Lorentzian response growing $0.23 \to 0.43 \to 0.70$ mV toward the Hopf ($1/\gamma$); above the Hopf, frequency dominance read by output frequency, without a phase-capture test.	The existence result: the beat is synthesized and amplified.	Figs. 4, 5; Fig. S4.
§4.3–4.4 Carrier independence, kHz roll-off, curvature law	Flat response over 25–300 Hz at fixed polarization; $1/f_c^2$ roll-off for 16 ms, retained access for 0.2 ms; two tones at 2 kHz with 2.2×10^{-13} mV input spectral residue. Curvature law to 0.9%; slope 1.99; linearization collapses the response 634×; two-tone/AM ratio 1.0000 at 0.02 mV and 0.988 at 1 mV.	Verifies the static detector predictions. Retain the one-point open-loop scope of the surrogate check; recurrent AM sweeps are explicitly qualified.	Figs. 6, 7; Figs. S6, S9; Esmaeilpour 2020, 2021.
§4.5 Realistic-head maps	From 1:1 to 1:3, the envelope focus moves 2.5 mm anterior, the mean-square focus 4.5 mm posterior, and the quadratic and isotropic beat maps rescale by 0.75 without moving. The directions persist in all 32 sensitivity combinations; magnitudes depend on orientation and region definitions. At 1:1 the envelope contrast is 1.22, versus 1.08 for the square-root quadratic map. Retain the absolute means, sweep, sensitivity ranges, and the distinction between a geometric mean-square map and a signed physiological response.	Gives the spatial consequence of the detector law on one realistic anatomy. Current-ratio steering separates the geometric maps; physiological detection remains to be measured. The following coupling sweep isolates recurrent gain.	Figure 8; Ricardo Salvador's FEM contribution, PR 1 (`d779094`).

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§4.6 Coupling sets the gain at fixed detection	Pinned JR coupling curve: $\sim$ 30$\times$ growth, $1/\gamma$ then saturation; the same gain–γ trend via p; stable-focus side only. Frequency and modal projections vary too.	The mesoscale claim: network gain grows near a weakly damped mode while the JR local curvature is fixed; parameter paths need not have identical finite-distance gain.	Fig. 9; Fig. S3; Clusella 2023.
§4.7 Timing modulation exceeds the mean-rate shift	With two tones, modulation grows 13.2$\times$ (26 to 339 mHz) against a mean shift of 4.4$\times$ ($-$1.8 to $-$7.7 mHz); ratio 14.5 far from the Hopf, peaking at 58 and falling to 44 nearest the Hopf. Retain the surrogate comparison (19, 66, 45) and rectification of the amplified oscillation: finite $G(0)$ does not bound this extra mean shift.	Shows when resonant timing modulation exceeds the mean shift and where that ratio turns over; retain finite-amplitude rectification and both waveform controls.	Fig. S11; Vieira 2024.
§4.8 TI and tACS share the linear amplifier	The TI/tACS ratio cancels the common transfer in linear response (0.143 predicted); the surrogate ratio rises to 0.24 and 0.47 near the Hopf (two tones: 0.24 and 0.46). Retain the unequal-input and field-free-pinning qualifications; matched-source tACS follows TI with ratios 0.99, 0.94, 0.92 for the surrogate and 0.99, 0.93, 0.90 for two tones at three tested couplings. Both driven means can shift. Do not infer a universal efficiency gain.	Separates the shared-amplifier result from the coupling-sweep protocol: resonance gives no TI advantage in linear response; stronger finite-amplitude responses saturate first.	Fig. S3; the three matched-source controls.
§4.9 Exact mean-field confirmation	PING generates and amplifies the beat through v_E^2; diagonal ridge with operating point; the fast-carrier source predicts the second-order response within 1.1% at four tested stable states, and within 0.1%/0.03% at 1/2 kHz; this does not bound source variation across every sweep. NMM2 has two competing modes: the critical mode dominates at low coupling and near the Hopf, the 55 Hz mode in between. The near-Hopf linear exponent is 1.02 over a 5.8-fold damping range; the full-sweep exponent is 0.71. Retain finite-amplitude departures and the $A_f = 2$ check. The 50.5 Hz endpoint remains attracting for both waveforms despite the DC-only screen; endpoint diagnostics are in Methods. The peak figure is remeasured on a finer grid.	Confirms detection and amplification in a neuron-derived mean field; the dominant mode can change across a coupling sweep, so the local near-Hopf law has a stated range.	Figs. 10, 11; Figs. S8, S10; Clusella 2023.

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§4.10 QIF responses separate beat modulation from induced gamma	QIF: at $\bar{\eta} = 0.85$, stimulation induces gamma through a rectified E-drive shift beyond the E-only Hopf; at 1.5, gamma dominates a smaller beat component. Phase folding suppresses that gamma. Retain sidebands, beat depths 0.29/0.23, rate +8–11%, and weak per-neuron locking; these runs do not test weak-input scaling or establish gating. At $A_f = 10$, the stable-state spiking depth is 0.093 with rate +2.8%; retain the cutoff sensitivity (details now in Methods): depths 0.093/0.127/0.139 at spike cutoffs 100/200/400, versus mean-field 0.128. At half step, the higher-cutoff depths are 0.121/0.141; cutoff dependence persists and quantitative convergence remains unestablished. In the equal-RMS mean-field comparison, both waveforms induce gamma at the larger amplitude; TI adds a resolved beat component. A fourfold longer window suppresses the carrier's apparent 55 Hz line below 10^{-4}. The complementary comparison is Figure 13 (formerly S13); retain the original threshold-crossing and autonomous-gamma examples in Fig. 12 and S12.	Larger modulation than mean-rate change at mesoscale and microscale, qualitatively consistent with Vieira; retain the finite-amplitude scope, nonzero mean shift, and weak per-neuron locking.	Figs. 12–13; Figs. S11–S12; Vieira 2024.

LaNMM, laminar neural mass model; tACS, transcranial alternating-current stimulation; SNIC, saddle-node on an invariant circle; RK4, fourth-order Runge–Kutta.

Interpretation and supporting material

Third block: Sections 5–6 and Appendices A–G

The Discussion changes level from model results to physiology and states each scope limit at the claim it limits. The appendices hold the notation, the model equations, the two bounds, the laminar extension, and the metrics.

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§5 Discussion	Weak-field cellular contributions enter an effective quadratic response; QIF provides a realization, without requiring channel identification. Weak amplitude and fast relaxation justify distinct approximations. Access candidates include nodes and AIS; strong terminal polarization alone does not establish fast charging (RC independent of area). Cifra excludes fields below 0.1 kV/m from the review's scope, not by a universal threshold. Cohen's periaxonal space ($\sim$ 1 ms fitted decay), Luff's 1 ms RC, Semenov's 0.7 μs charging: a 10–100$\times$ shorter time is a hypothesis. Dose map: excursion versus product and the DC term, referencing the derivation in §2.5–2.6. The equal-RMS carrier comparison separates the operating-point shift from the beat. Violante's 1:3 versus 1:1 currents predict a 3/4 quadratic beat at fixed state, with a spatially changing mean-square map. Their direct BOLD contrast ($P = 0.015$) has no stimulation-by-region interaction ($P = 0.76$); envelope steering has model and cadaver support, but BOLD and functional connectivity do not measure the neural beat map. Intensity changes the mechanism. Shared amplifier with tACS; temporal, not spatial, criticality. Couplings 0.12–0.5 mV per V/m; $\mathcal{R}_{\mathrm{det}} \simeq |\Sigma''|\varepsilon/4\Sigma'$; $\mathcal{Q} \sim 10$. Vieira's unmatched-current two-pair TI result (0.34, CI 0.04–0.65) is separate from the matched-depth AM control (PLV 0.10 versus 0.20). The illustrative 0.1 mV budget gives 0.8% at $v^* = 8.30$ mV; drive and PLV are different observables. The numerical 0.348 ceiling extends by positive step mixtures to bounded nondecreasing memoryless transfers, for the carrier-averaged, matched-peak, open-loop comparison. Larger common matched polarization does not evade it. Limitations: Plovie 2025, noise, columns, cascaded synapses cannot solve access, mixed human results.	Reinterprets each result in physiological terms and places every caveat where it bites. Retain each level (anatomical access, mixing, time scales, dosimetry, intensity, efficiency, scale, data, bound, limits), with access preceding detection. Retain matched-peak versus equal-RMS distinctions, and matching of input coupling, readout, and state for the shared amplifier.	Fig. S3 (tACS); Cohen 2020; Luff 2024 (PWM-TI); Semenov 2025; Vieira 2024; App. D.

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§5.1 Falsifiable predictions	Carrier sweeps with measured local polarization; independent-amplitude tests of product versus minimum at fixed weaker tone; current-ratio steering with fixed pairs and projection directions, normalized spatial beat maps, and DC/state controls. For $2 + 2$ versus $1 + 3$ mA, envelope-excursion factors range from 0.5 to 1.5, the quadratic-beat factor is 0.75 everywhere, and the mean-square factor is 1.25 at equal per-current magnitudes, tending to 0.25 or 2.25 when the first or second carrier dominates, respectively. This geometric factor does not assign excitation or suppression. Retain the tests of state and carrier dependence; amplitude slope and harmonic ratio off resonance; beat-frequency selectivity; timing versus rate; synaptic blockade; equal-power carrier controls. The independent-amplitude test reconstructs the slow excursion, which is not a single beat lock-in. Equal-RMS evidence is from the PING mean field, with spiking-network and in vivo controls still needed. Table 4 preserves network-gain calibration, weak-input conditions, and existing carrier comparisons.	Converts the mechanism into falsifiable statements within the Discussion; distinguishes existing carrier sweeps from an access-corrected test.	Table 4; Vieira 2024; Caldas-Martinez 2024; Opancar 2025.
§6 Conclusion	Product law and DC map; shared linear amplifier; timing over mean-rate change; field-induced gamma versus beat. Retain the calibrated-access requirement, static-versus-dynamic condition, and one-filter accounting. The equal-power carrier control is a mean-field result, and a large output alone does not identify its mixer.	Restates the headline results with their scope and measurement requirements.	—
App. A–B Notation and units; model equations	Table 5 (symbols, meaning, units, and definition locations; shared with the supplement and included in the JNE article); JR equations (Eq. 26) and Table 6. App. B.1 now holds the compartment matrix with SI units and the passive-cable estimate; B.2 holds the single-cell rate and dynamic MPR Jacobian/phasor derivation.	Recovery aids; definitions remain local.	—
App. C–D Cubic normal form; efficiency bound	Stuart–Landau has no second-order source (Eq. 30); numerical control at the floor. $\mathcal{R}_{\mathrm{det}} \leq \rho\varepsilon/4$ (Eq. 31); family supremum 0.3479 at $\delta/2\varepsilon = 0.523$ (Eq. 33), a step detector with harmonic ratio 0.151; the microvolt width is an idealization, and the maximum is numerical, not certified. The nonnegative direct coefficient makes the positive-step-mixture extension valid.	C supports §2.7 (susceptibility is not detection); D supports §2.4 and the Discussion's efficiency paragraphs.	`threshold_bound_check.py`.

Place in v0.15.0-draft20	Scientific content to retain	Contribution to the argument	Examples / links
App. E.1–E.2 Laminar methods and low-carrier results	Method: field as a signed polarization inside the sigmoid (Eq. 34), with no rate clipping; a clip would itself rectify. RK4 integration, alpha-band power including autonomous activity, signed power maps, and the 17-beat lock-in mean with 3/10 s windows and field-free background controls. Results: frequency selection at 25–80 Hz carriers up to 3 mV, outside the perturbative regime; 95%/97% of P_1-driven grids below baseline. Off-alpha mean lock-in peaks at 39 Hz for superficial drive and 25 Hz for direct drive, with unequal amplitudes. Retain the 3/10 s comparison, field-free background, and limits of the near-alpha readout. The old 7%/25% claims are replaced by the reproducible carrier-profile table.	Places the extension's method beside its results. A band-limited intermediary makes the carrier choice matter; these results do not carry the kHz argument.	Fig. S1, S2, S7; Gordon 2019.
App. F–G Vocabulary metrics; supplementary figures	Dominance screen versus capture test; folded-rate metric versus gamma-amplitude gating; frequency-dominance protocol. Figures S2–S12 support §4.2–4.10 and the Discussion; S13 compares the geometric maps derived in §2.5–2.6.	Metrics behind the three terms; the figure bank for the JNE supplement.	Figs. S2–S13.

Placement decisions through 9 October 2026. The map above records the approved section and method locations. LaNMM methods remain beside their results in Appendix E; the equal-RMS control remains in the main article, and S3, S10, and S11 remain supplementary. Existing equations, figures, examples, numerical results, and the QIF convergence qualification are retained. The abstract's mean-map clause states the boundary consequence for the generated drive with its uniformity conditions. The reading baseline is draft13 (`c620335`). Coauthor confirmation, remaining funding declarations and the seven retained CUT? passages remain pending. Draft20 fills the approved FEM locations with Ricardo's methods, results, and Figure 8 from private-mirror PR 1. No earlier result or example is removed. The shared abstract is unchanged; whether to include a FEM finding there is Giulio's decision. Draft19 distinguished geometric intensity, signed source, and circuit response and was shared privately with Ricardo on 14 October. The public GitHub push, Calliope and Zenodo releases, and circulation to other coauthors remain held; see `todo.md`.